

Existence of market equilibrium

A market does not always achieve an equilibrium. Consider the following TPM:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \end{pmatrix}.$$

Then

$$P^{(2k)} = \begin{pmatrix} 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \end{pmatrix}$$

and

$$P^{(2k+1)} = \begin{pmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \end{pmatrix}$$

where k is a positive integer. A market that follows this model, does not reach an equilibrium.

Classification of states

Accessible: State j is accessible from state i if for some $n \geq 0$ we have

$$p_{ij}^{(n)} > 0$$

We then write, $i \rightarrow j$.

Properties:

1. $i \rightarrow i$ (since $p_{ii}^{(0)} = 1$).
2. If $i \rightarrow j$ and $j \rightarrow k$, then $i \rightarrow k$.

Exercise:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \end{pmatrix}$$

Is $1 \rightarrow 3$?

Solution:

$1 \rightarrow 2$ and $2 \rightarrow 3$. Therefore, $1 \rightarrow 3$.

Communicating:

States i and j communicate if $i \rightarrow j$ and $j \rightarrow i$. We then write $i \leftrightarrow j$.

Properties

Reflexivity: $i \leftrightarrow i$ (since $p_{ii}^{(0)} = 1$).

Symmetry: If $i \leftrightarrow j$, then $j \leftrightarrow i$.

Transitivity: If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.

Exercise:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 1 & 0 \end{pmatrix}$$

Is $1 \leftrightarrow 3$?

Solution:

$1 \rightarrow 2$ and $2 \rightarrow 3$. Therefore, $1 \rightarrow 3$. Similarly, $3 \rightarrow 1$. Therefore, $1 \leftrightarrow 3$.

Communicating class (Equivalence class)

A set of all states that communicate with each other is called an equivalence class.

A communicating class is closed if the process cannot leave the class. That is, a class C is closed if $i \in C$ and $j \notin C$ implies $i \nrightarrow j$.

Irreducible Markov Chain

A Markov Chain is called irreducible if all its states belong to one equivalence class. Otherwise, it is called reducible.

Exercise:

$$\begin{array}{c} \begin{matrix} & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{pmatrix} 0.2 & 0.3 & 0.4 & 0.1 \\ 0.3 & 0.3 & 0.3 & 0.1 \\ 0.1 & 0.2 & 0.2 & 0.5 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

Is the M.C. irreducible?

Solution:

Here, $1 \leftrightarrow 2 \leftrightarrow 3$. Though 4 is accessible from these states, none of these states is accessible from 4. Therefore, the M.C. is reducible.

[In fact, 4 is an *absorbing state*. Once the process is in state 4, it cannot leave 4.]

Exercise:

$$\begin{array}{c} \begin{matrix} & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{pmatrix} 0.4 & 0 & 0.3 & 0.3 \\ 0.5 & 0.2 & 0.3 & 0 \\ 0.2 & 0.2 & 0.4 & 0.2 \\ 0.3 & 0.3 & 0.3 & 0.1 \end{pmatrix} \end{array}$$

Is the M.C. irreducible?

Solution:

Here, $1 \leftrightarrow 2$, $2 \leftrightarrow 3$ and $3 \leftrightarrow 4$. Thus, $1 \leftrightarrow 2 \leftrightarrow 3 \leftrightarrow 4$. Therefore, the M.C. is irreducible.

Periodic state

A state i is said to be periodic with period d if d is the largest integer such that $p_{ii}^{(n)} > 0 \Rightarrow n$ is a multiple of d .

Example:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Is the chain irreducible? Determine the period of state 1.

Solution:

Here, $1 \leftrightarrow 2 \leftrightarrow 3$. Therefore, the chain is irreducible. The set of integers n for which $p_{11}^{(n)} > 0$ is $\{3, 6, 9, \dots\}$. Therefore, $d = 3$.

Theorem

If $i \leftrightarrow j$, then i and j have the same period.

Example:

In the above example, all the states are communicating. So, each state has period 3.

Example:

Let the set of positive integers n for which $p_{11}^{(n)} > 0$ be $\{3, 7, 9, \dots\}$. Then, $d = 1$, because the Greatest Common Divisor (GCD) is 1.

Note:

When period $d = 1$, we say that the state is aperiodic.

Exercise:

$$\begin{matrix} & 1 & 2 \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 0.2 & 0.8 \\ 0.3 & 0.7 \end{pmatrix} \end{matrix}$$

Determine the period of each state.

Solution:

For state 1: The set of integers n for which $p_{11}^{(n)} > 0$ is $\{1, \dots\}$. Therefore, $d = 1$. That is, state 1 is aperiodic.

Since $1 \leftrightarrow 2$, state 2 is also aperiodic.

Exercise:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 1/3 & 2/3 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/4 & 3/4 \end{pmatrix} \end{matrix}.$$

Is the chain irreducible? Determine the period of each state.

Solution:

$1 \leftrightarrow 2 \leftrightarrow 3$. Therefore, the chain is irreducible.

State 1 has period 1. Since all the states are communicating, each has period 1.

Exercise:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 2/3 & 1/3 \\ 0 & 0 & 1/4 & 3/4 \end{pmatrix} \end{matrix}$$

Is the M.C. irreducible? Determine the period of each state.

Solution:

$1 \leftrightarrow 2$. Again, $3 \leftrightarrow 4$. However, $\{1, 2\}$ and $\{3, 4\}$ are two equivalence classes that do not communicate with each other. The chain is reducible.

State 1 (and, therefore, state 2) has period 1. State 3 (and, therefore, state 4) has period 1.

Exercise:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{pmatrix} 1/2 & 0 & 0 & 0 & 0 & 1/2 \\ 0 & 1/3 & 0 & 0 & 2/3 & 0 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2/3 & 0 & 0 & 1/3 & 0 \\ 1/2 & 0 & 0 & 0 & 0 & 1/2 \end{pmatrix} \end{matrix}$$

Determine the equivalence class(es) and the period of each state.

Solution:

$1 \leftrightarrow 6$. $\{1, 6\}$ is a closed class.

$2 \leftrightarrow 5$. $\{2, 5\}$ is a closed class.

$\{4\}$ is a closed class (diagonal element $p_{44} = 1$). State 4 is an absorbing state.

$\{3\}$ is a transient (temporary) class. Once the process leaves this class, it can never come back.

$\{1, 6\}$ has period 1. $\{2, 5\}$ has period 1. $\{4\}$ has period 1. Period of $\{3\}$, a transient class, is irrelevant.

Exercise:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix} \end{matrix}$$

Determine the equivalence class(es) and the period of each state.

Solution:

Do it yourself.

Recurrent and transient states

Identification of recurrent and transient states helps in the study of the limiting behavior of DTMC.

Let

$$f_{ii}^{(n)} = P(X_n = i, X_1 \neq i, \dots, X_{n-1} \neq i \mid X_0 = i)$$

= Probability that the first recurrence to i is at the n th step

Then, the recurrence probability is given as

$$f_i = f_{ii} = \sum_{n=1}^{\infty} f_{ii}^{(n)}$$

A state i is recurrent if $f_i = 1$ (return is certain).

A state i is transient if $f_i < 1$ (return is NOT certain).

Example:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix} \end{matrix}$$

$$f_{11}^{(1)} = p_{11} = 0.7$$

$$f_{11}^{(2)} = 0.3 \times 0.2$$

$$f_{11}^{(3)} = 0.3 \times 0.8 \times 0.2$$

$$f_{11}^{(4)} = 0.3 \times 0.8^2 \times 0.2$$

$$f_{11}^{(5)} = 0.3 \times 0.8^3 \times 0.2$$

and so on. Therefore,

$$\begin{aligned} f_1 &= f_{11}^{(1)} + f_{11}^{(2)} + f_{11}^{(3)} + f_{11}^{(4)} + \dots \\ &= 0.7 + 0.3 \times 0.2 (1 + 0.8 + 0.8^2 + \dots) \\ &= 0.7 + 0.3 \\ &= 1 \end{aligned}$$

Therefore, state 1 is recurrent. Similarly, state 2 is also recurrent.

Example:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0.7 & 0.3 & 0 \\ 0 & 0.4 & 0.6 \\ 0 & 0.2 & 0.8 \end{pmatrix} \end{matrix}$$

$$f_{11}^{(1)} = p_{11} = 0.7$$

$$f_{11}^{(2)} = 0.3 \times 0 = 0$$

$$f_{11}^{(3)} = 0.3 \times 0.4 \times 0 + 0.3 \times 0.6 \times 0 = 0$$

and so on. Therefore, $f_1 = 0.7 < 1$. Thus, state 1 is transient.

Mean recurrence time

Let

T_i = Time for first recurrence to state i (given $X_0 = i$)

Then, the mean recurrence time is given as

$$m_i = E(T_i | X_0 = i)$$

$$= \sum_{n=1}^{\infty} n f_{ii}^{(n)}$$

Example:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix} \end{matrix}$$

Calculate the mean recurrence time of state 1.

Solution:

Mean recurrence time of state 1:

$$\begin{aligned} m_1 &= \sum_{n=1}^{\infty} n f_{ii}^{(n)} \\ &= 1 \times 0.7 + 2 \times 0.3 \times 0.2 + 3 \times 0.3 \times 0.8 \times 0.2 + 4 \times 0.3 \times 0.8^2 \times 0.2 + \dots \\ &= 0.7 + 0.3 \times 0.2 (2 + 3 \times 0.8 + 4 \times 0.8^2 + \dots) \\ &= 2.5 \end{aligned}$$

- A recurrent state i is said to be positive recurrent if $m_i < \infty$ and null recurrent if $m_i = \infty$.
- For transient states, mean recurrence time is necessarily infinite.
- Recurrence is a class property. That is, an equivalence class is positive recurrent or null recurrent or transient.
- In a finite state-space Markov Chain, there is no null recurrent state.
- In a finite state-space Markov Chain, all states cannot be transient. That is, at least one state will be recurrent (positive recurrent).
- Let C be a finite closed class. Then, all the states in C are positive recurrent.
- Let T be a finite class that is open. Then, all the states in T are transient.

Example:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0.7 & 0.3 \\ 0 & 0 & 0.2 & 0.8 \end{pmatrix} \end{matrix}$$

Classify the states of the chain.

Solution:

$\{1, 2\}$ and $\{3, 4\}$ are two equivalence classes.

$\{1, 2\}$ is a finite closed class and, therefore, is positive recurrent. Similarly, $\{3, 4\}$ is also positive recurrent.

Example:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{pmatrix} 1/2 & 0 & 0 & 0 & 0 & 1/2 \\ 0 & 1/3 & 0 & 0 & 2/3 & 0 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2/3 & 0 & 0 & 1/3 & 0 \\ 1/2 & 0 & 0 & 0 & 0 & 1/2 \end{pmatrix} \end{matrix}$$

Classify the states of the chain.

Solution:

The equivalence classes are $\{1, 6\}$, $\{2, 5\}$, $\{3\}$ and $\{4\}$.

$\{1, 6\}$ is a finite closed class and, therefore, is positive recurrent. Similarly, $\{2, 5\}$ and $\{4\}$ are positive recurrent classes. $\{3\}$ is a transient class.

Ergodic states

A positive recurrent and aperiodic state is called ergodic. If a Markov chain is irreducible and the states are ergodic, then the chain is called an ergodic chain.

Example:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix} \end{matrix}$$

The chain is finite and irreducible. All the states are positive recurrent. Period of each state is 1. Thus, all the states are ergodic. Therefore, the chain is ergodic.

Hidden Markov Models

A Hidden Markov Model (HMM) consists of two distinct stochastic processes: an invisible (hidden) process X_n and an observable process Y_n . The hidden process is a Markov Chain. Also

$$P(Y_n = j \mid Y_{n-1} = j_{n-1}, X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(Y_n = j \mid X_n = i)$$

That is, past values of X and Y does not influence the current value of Y given the current value of X . Then, (X_n, Y_n) is an HMM. We often want to learn about the state of X by observing Y .

Example

You want to determine whether your friend is happy (H) or unhappy (U). The only information you can obtain is by asking him what he is doing. He may say: reading book (B), drinking coffee (C) or walking (W).

The mental state of your friend forms Markov Chain with two states, H and U, but you cannot observe it directly; the process is *hidden*. On each day, the chance that he says B, C or W depending only on his mental condition on that day.

The *observations* (B, C, W) along with the *hidden* states (H, U) form a hidden Markov model (HMM). From past experience, the probabilities of this model have been estimated as:

Your initial belief:

$$\mathbf{a} = (H, U) = (0.6, 0.4)$$

Transition Probability Matrix:

$$P = \begin{matrix} & \begin{matrix} H & U \end{matrix} \\ \begin{matrix} H \\ U \end{matrix} & \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix} \end{matrix}$$

Observation Likelihood Matrix:

$$\begin{matrix} & \begin{matrix} B & C & W \end{matrix} \\ \begin{matrix} H \\ U \end{matrix} & \begin{pmatrix} 0.1 & 0.5 & 0.4 \\ 0.3 & 0.5 & 0.2 \end{pmatrix} \end{matrix}$$

Suppose, two consecutive days' observations are C and B. We want to choose between 4 possible mental state sequences: HH, HU, UH and UU. We calculate the probability of HU (say) as follows:

$$\begin{aligned}P(HU \cap CB) &= P(X_1 = H)P(X_2 = U|X_1 = H)P(Y_1 = C|X_1 = H)P(Y_2 = B|X_2 = U) \\&= 0.6 \times 0.3 \times 0.5 \times 0.3 \\&= 0.027\end{aligned}$$

The sequence with the highest calculated probability is chosen.

For three consecutive days' observations C, B and B (say), we want to choose between 8 possible mental state sequences: HHH, HHU, HUH, HUU, UHH, UHU, UUH and UUU. To deal with the exponentially increasing number of possible sequences, the Viterbi Algorithm is often used.